PCB Design Review Checklist

DIELCHA DIGILAB | Comprehensive Schematic, Layout & Manufacturing Review

A structured framework for conducting peer or self-reviews of printed circuit board designs, covering schematic intent, component selection, signal integrity, power distribution, and thermal performance.

1. Schematic & Functional Design Intent

Review Parameter	Status
Block diagrams and hierarchical sheets properly structured and labeled.	Check
Power supply topology verified (voltage levels, current draw, decoupling capacitors).	Check
Pull-up/pull-down resistors configured correctly on control lines, I2C buses, and reset pins.	Check
Unused IC pins terminated correctly (tied to VCC/GND or left floating per datasheet).	Check

2. Component Selection & Lifecycle

Review Parameter	Status
All active and passive parts verified for lifecycle status (avoiding Obsolete/NRND parts).	Check
Second-source options identified for critical ICs and passives.	Check
Package footprints match manufacturer recommended land patterns exactly.	Check
Voltage and power ratings derated appropriately (e.g., resistor wattage, capacitor voltage ratings).	Check

3. Layout, Routing & Signal Integrity

Review Parameter	Status
High-speed and clock traces kept short with continuous reference ground planes underneath.	Check
Analog and digital grounds separated or properly joined at a single point (star grounding).	Check
Power traces sized adequately for expected current densities (trace width calculator used).	Check
Via sizing and count sufficient for high-current power transitions and thermal dissipation.	Check

4. Thermal Management & DFM Readiness

Review Parameter	Status
Thermal reliefs used on planes connected through-hole pads to ensure solderability.	Check
Heat-generating components (regulators, MOSFETs, drivers) placed with adequate copper area.	Check
Test points added for crucial debugging and in-circuit testing (ICT).	Check

Assembly clearances checked for tall components and connectors.	Check